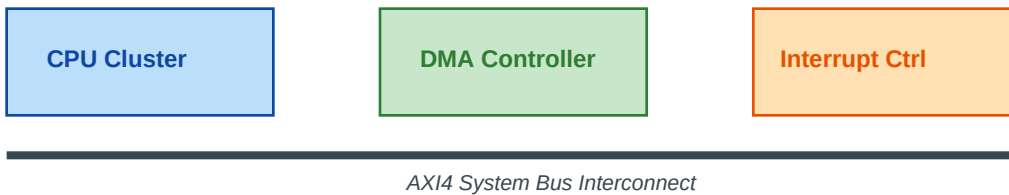


System-on-Chip (SoC) Architectural Specification

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1. Executive System Overview & Bus Interconnect

This specification defines the high-level hardware architecture for the multi-core SoC platform. The system integrates an ARM Cortex processor cluster, high-performance DMA controller, Vectored Interrupt Controller (VIC), and an AXI-to-APB peripheral bridge over an AXI4 system bus topology.



2. Direct Memory Access (DMA) Controller Subsystem

The DMA controller manages high-throughput scatter-gather data transfers between system memory and peripheral devices without CPU intervention. It supports 8 concurrent DMA channels, configurable burst sizes, and priority arbitration.

2.1 DMA Controller Register Map

Register Name	Offset	Access	Reset Value	Description
DMA_CTRL	0x00	R/W	0x0000_0000	Control Register (Bit 0: Enable, Bit 1: Interrupt Enable)
DMA_STATUS	0x04	RO	0x0000_0000	Channel Status Flags (Transfer Complete, Error)
DMA_SRC_ADDR	0x08	R/W	0x0000_0000	Source Base Memory Address
DMA_DST_ADDR	0x0C	R/W	0x0000_0000	Destination Base Memory Address
DMA_XFER_SIZE	0x10	R/W	0x0000_0000	Transfer Length in Bytes (Max 64KB)

3. Vectored Interrupt Controller (VIC)

The Vectored Interrupt Controller routes 32 hardware interrupt lines to the CPU core. It features programmable interrupt priority levels (0-15) and fast hardware interrupt vector lookup.

3.1 VIC Register Map

Register Name	Offset	Access	Reset Value	Description
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VIC_IRQ_STATUS	0x00	RO	0x0000_0000	Active Raw Interrupt Status
VIC_INT_ENABLE	0x04	R/W	0x0000_0000	Interrupt Enable Mask
VIC_VEC_ADDR	0x08	RO	0x0000_0000	Vector Address of Active Highest Priority IRQ